# CodePilot AI - Autonomous Software Engineering Agent

Bug report analysis, code retrieval, patch generation, validation checks, review-ready diff

| Project Type | Best For | Deliverables |
| --- | --- | --- |
| Backend / Distributed Systems Demo | Amazon SDE I resume + interview explanation | Runnable code, demo script, tests, PDFs |

## 1. Problem Statement

Developers spend time reading bug reports, locating files, generating fixes, running tests, and preparing changes for review.

## 2. Architecture Flow

Bug Report -> Repository Indexer -> Retrieval Engine -> Patch Generator -> Validation Runner -> Review Output -> Human Engineer

## 3. Core Features

- Indexes a sample repository for code search
- Retrieves relevant files based on bug report tokens
- Generates a unified code diff for a divide-by-zero bug
- Runs automated validation checks on patched code
- Returns review status for human approval
- Includes a browser demo and test suite

## 4. API Surface

| Endpoint | Purpose |
| --- | --- |
| POST /analyze | Analyze bug report and generate patch |
| GET /repo/files | List indexed repository files |
| GET / | Browser demo UI |

## 5. How to Run the Demo

Run these commands from the project folder:

```
pip install -r requirements.txt
uvicorn app.main:app --reload --port 8000
pytest -q
```

## 6. Demo Talking Points

I kept the system interview-friendly: retrieval, code indexing, patch generation, validation, and safe human review. The focus is the engineering workflow, not just an AI wrapper.

## 7. Resume Bullet

> Built an AI-powered software engineering agent that analyzes bug reports, retrieves relevant code, generates validated patches, runs automated checks, and prepares review-ready diffs.

## 8. What to Say in Interview

- Why I built it: to practice backend workflows that need correctness, reliability and scalability.
- Main challenge: keeping state transitions correct when retries or failures happen.
- What I learned: data structures, APIs, testing, failure handling, and clean modular design.
- Future scale-up: replace in-memory components with Redis/Kafka/SQS, add auth, metrics, tracing and cloud deployment.

## 9. Demo Output Snapshot

**POST /analyze**


```
{
"bug_report": "divide by zero crashes calculator",
"retrieved_context": [
{
"path": "calculator.py",
"score": 2,
"preview": "def add(a, b):\n return a + b\n\n\ndef divide(a, b):\n return a / b\n"
},
{
"path": "README.md",
"score": 2,
"preview": "# Sample Repository\n\nA tiny calculator package used by CodePilot AI to
demonstrate bug report analysis, code retrieval, patch generation, and test
validation.\n"
}
],
"generated_diff": "--- calculator.py\n+++ calculator.py\n@@ -3,4 +3,6 @@\n \n \n def
divide(a, b):\n+ if b == 0:\n+ return None\n return a / b\n",
"validation": {
"passed": true,
"checks": [
```